



PORTFOLIO

Aldiansyah Anugrah Ramadhan

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github.com/Aldiansyah-ar



About Me

Master's graduate in Computational Science from Bandung Institute of Technology (ITB), with a thesis focused on computational fluid dynamics simulation using Python. I have strong interest in AI, data engineering and software engineering. I also hold a Bachelor's degree in Physics from ITB, with a final project focused on gravitational waves analysis.

My expertise encompasses on AI model development, data engineering, and software engineering, utilizing tools such as Python, SQL and NoSQL databases, and JavaScript. As a Data Scientist, I have hands-on experience building data pipelines and integrating AI models into web-based applications.

With strong analytical approach, I am currently seeking opportunity where I can apply my expertise while further enhancing my skills. My goals are to advance my knowledge in AI model development and solve complex problems by leveraging my technical background.

Education



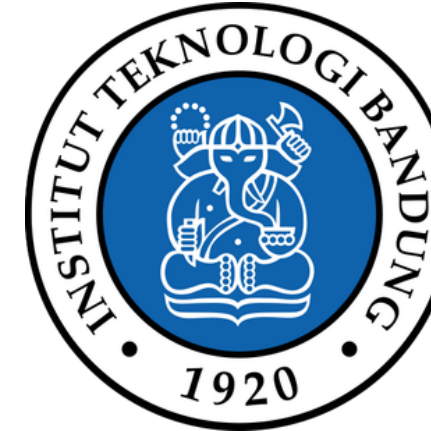
**Bandung Institute of Technology
Indonesia
Bachelor of Sciences
Physics Major
2016-2021**

Relevant Course:

Mathematical Methods in Physics, Modern Physics, Special Relativity, General Relativity, Linear Algebra.

Final Project:

Gravitational Waves in The $F(R)$ Gravity Theory Framework.



**Bandung Institute of Technology
Indonesia
Master of Sciences
Computational Science Major
2025-2026**

Relevant Course:

Computational Science Methods, Python Programming, Numerical Analysis, Algorithm and Programming in Science.

Thesis:

Development of The Finite Difference Method for Vorticity Flow Simulation.

Thesis Repository:

github.com/Aldiansyah-ar/Thesis

Professional Experiences



Al-Bahjah Islamic Senior High School
Cirebon, Indonesia
Physics Teacher
Apr 2022 - Jun 2023

- Developed a physics curriculum for senior high school students, focusing on analytical and conceptual mastery.



PT. Steradian Data Optima
Jakarta, Indonesia
Data Scientist
Aug 2023 - Jan 2024

- Built the robust backend infrastructure of the web application leveraging Django.
- Integrated machine learning models to power the Land Cover Classification web application.
- Automated data workflows by developing Python ETL pipelines seamlessly connected to the web application.

Other Experiences



Korea-ASEAN Digital Academy (KADA)
Cikarang, Indonesia
Participant
Jun 2025 - Aug 2025

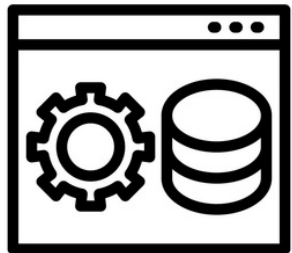
- Completed 250 hours of intensive training covering Frontend Development (HTML, CSS, React), Backend Development (Node.js, Express.js, MongoDB), Basics of Data Analysis, Cloud Service Utilization (AWS), DevOps (CI/CD, Containerization), and UI/UX.
- Developed a web application for AI-powered interview preparation using the MERN stack, integrating the Gemini API to dynamically generate IT-related interview questions, perform resume screening, enable live coding sessions, and provide model answers.



Bandung Institute of Technology
Bandung, Indonesia
Titian Karir Terpadu Assistant
Apr 2025 - Apr 2026

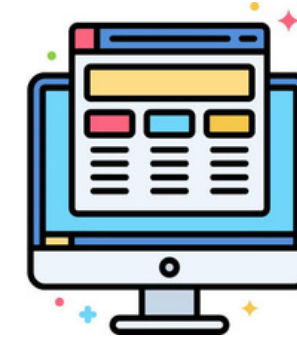
- Managed visitor flow and maintained site security to ensure a safe and seamless experience during the event.
- Served as the primary point of contact for Garena Indonesia, delivering company overviews and effectively resolving participant inquiries.
- Facilitated the end-to-end administration of corporate written selection tests, ensuring a smooth and structured examination process.

Skills



Backend:

Python (Django, FastAPI), Node.js (Express.js)



Frontend:

HTML, Tailwind CSS, React, Typescript



Databases:

SQL (PostgreSQL, MySQL), MongoDB.



DevOps, Cloud, & Tools:

Git, Linux, AWS, Postman, Docker



AI Engineering:

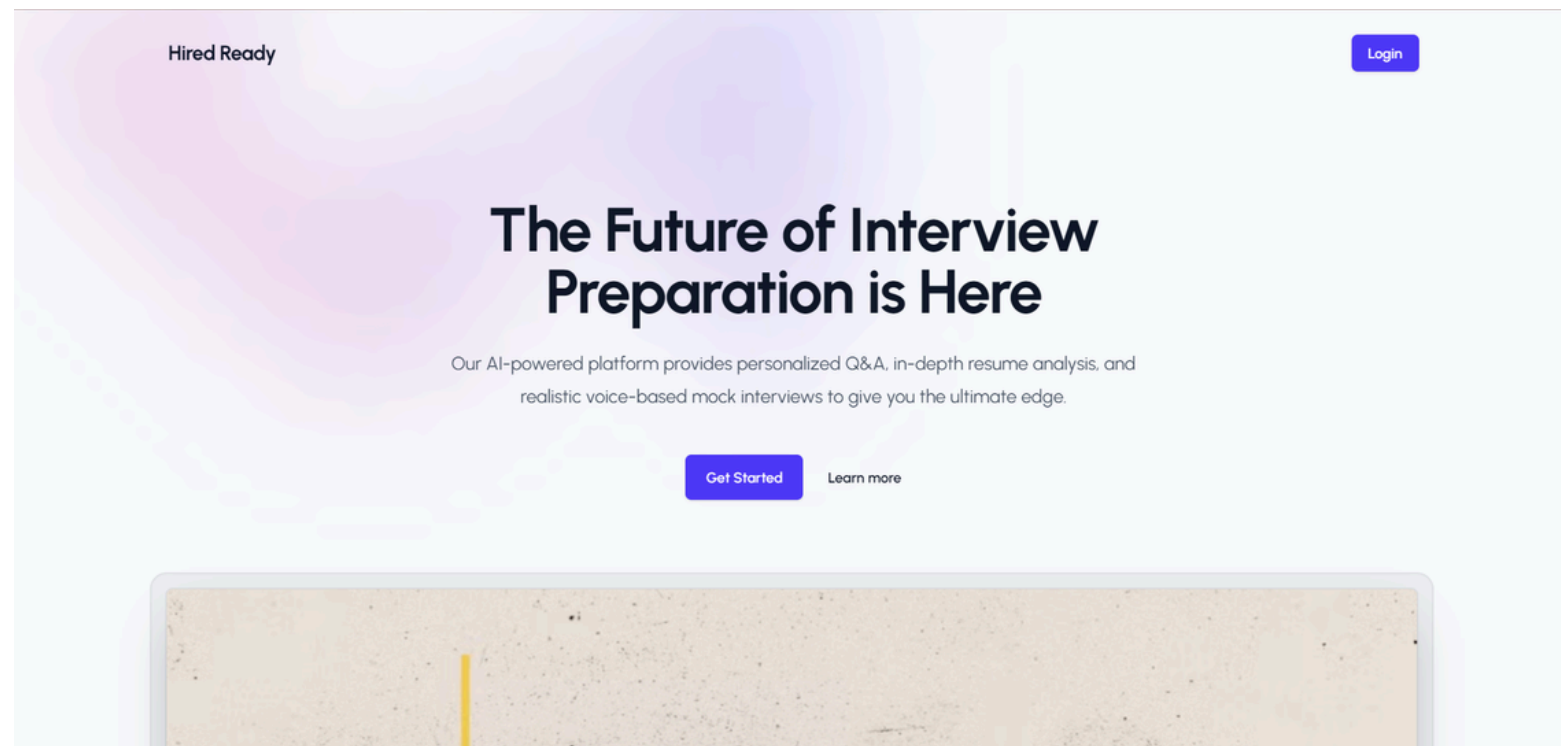
Machine Learning (Scikitlearn), Deep Learning (Tensorflow, PyTorch), LLM (Transformer, Gemini API), RAG

Projects

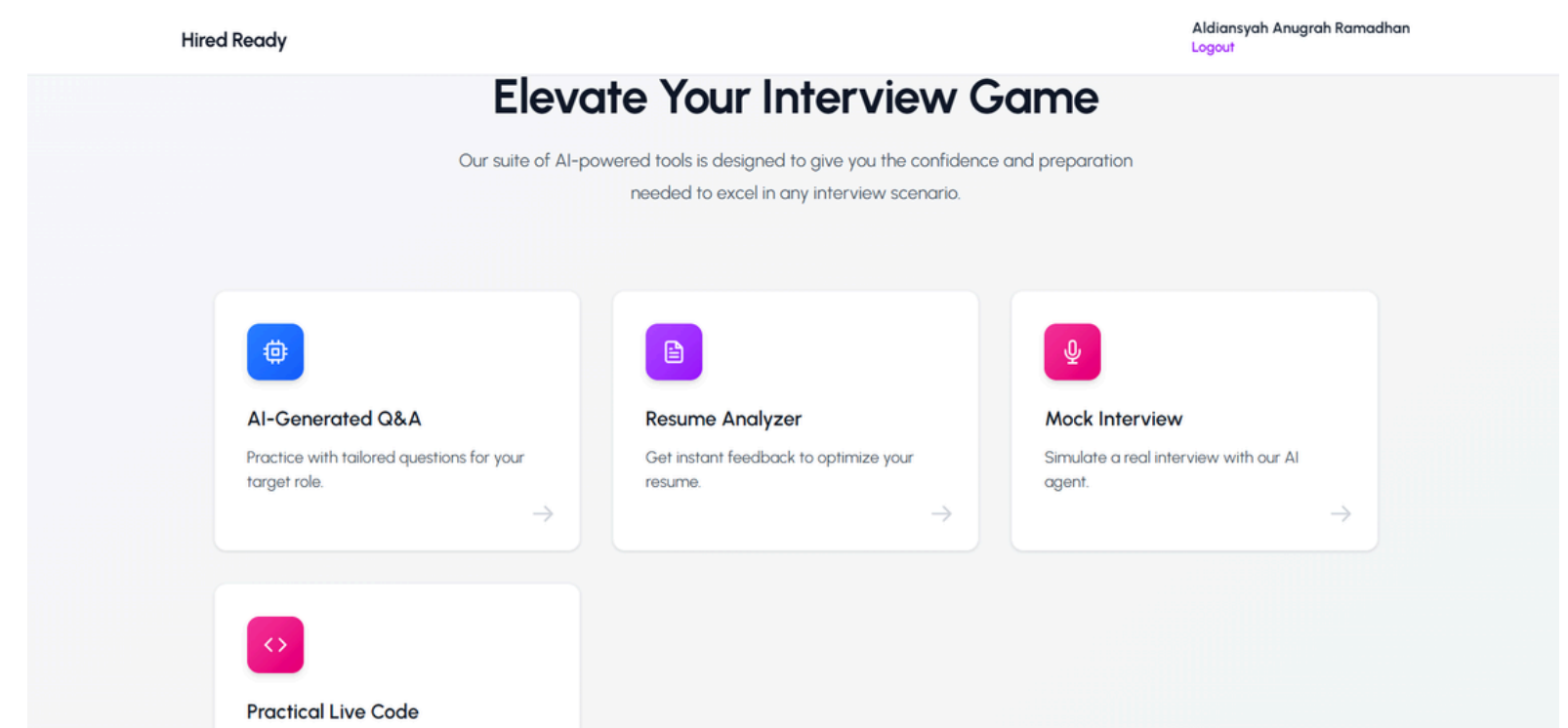
Hired Ready (KADA Capstone Project)

github.com/Aldiansyah-ar/HiredReady

An AI-powered web platform built with the MERN stack (MongoDB, Express.js, React, Node.js) and integrated with Gemini API. It help users prepare for technical and conceptual interviews through smart features such as dynamic IT question generation, resume screening and insights, voice-based mock interviews, and real-time live coding sessions with model answers.



Main page



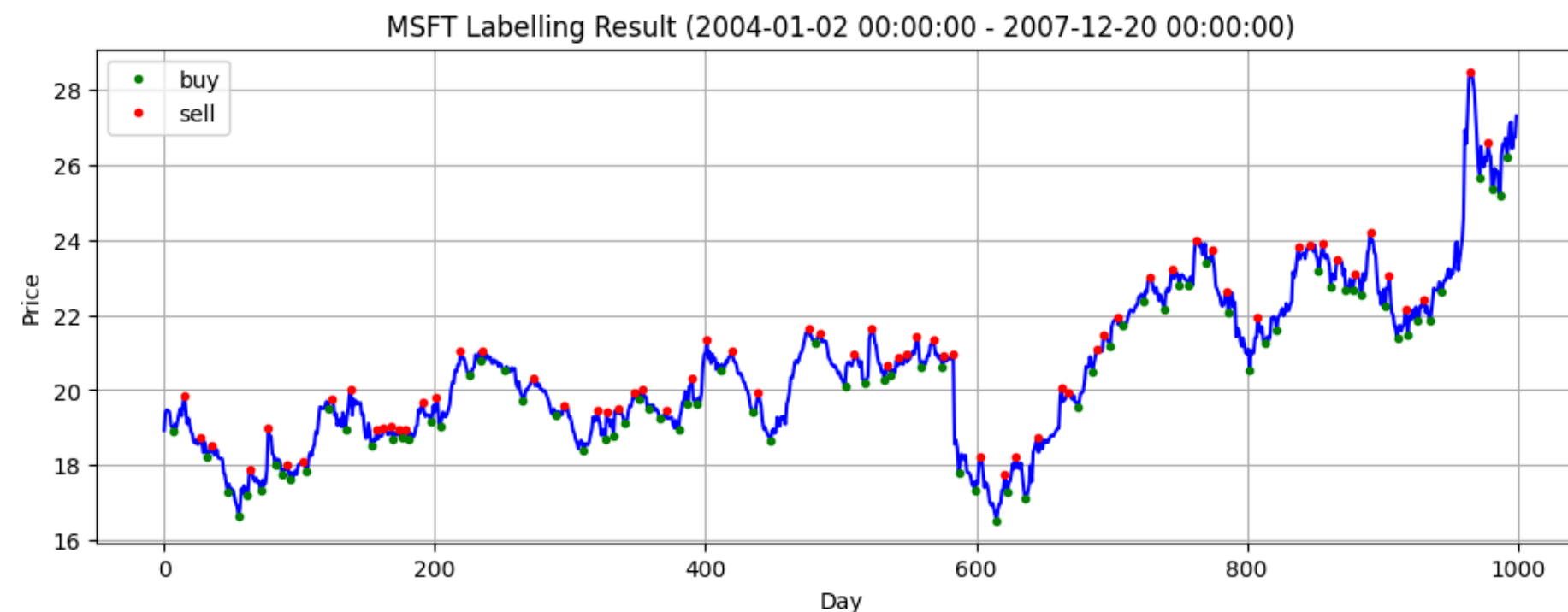
Menu

Projects

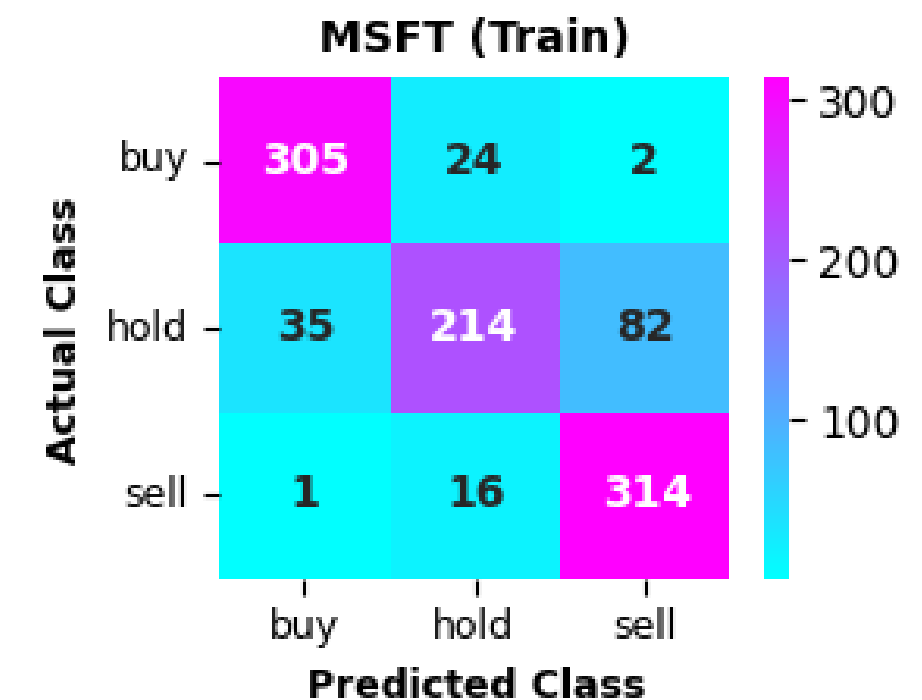
Classification of Buy, Hold, and Sell Points for US Electronics Company Stocks Using Convolutional Neural Networks

github.com/Aldiansyah-ar/BuyHoldSellPointClassification

Investor decisions in stock management rely on identifying optimal buy, hold, and sell points to determine the best asset strategy and minimize risk. Stock data from technology companies including Apple Inc., HP Inc., and Microsoft were transformed into image formats to enable the classification of buy, hold, and sell points using the CNN algorithm.



MSFT labelling result



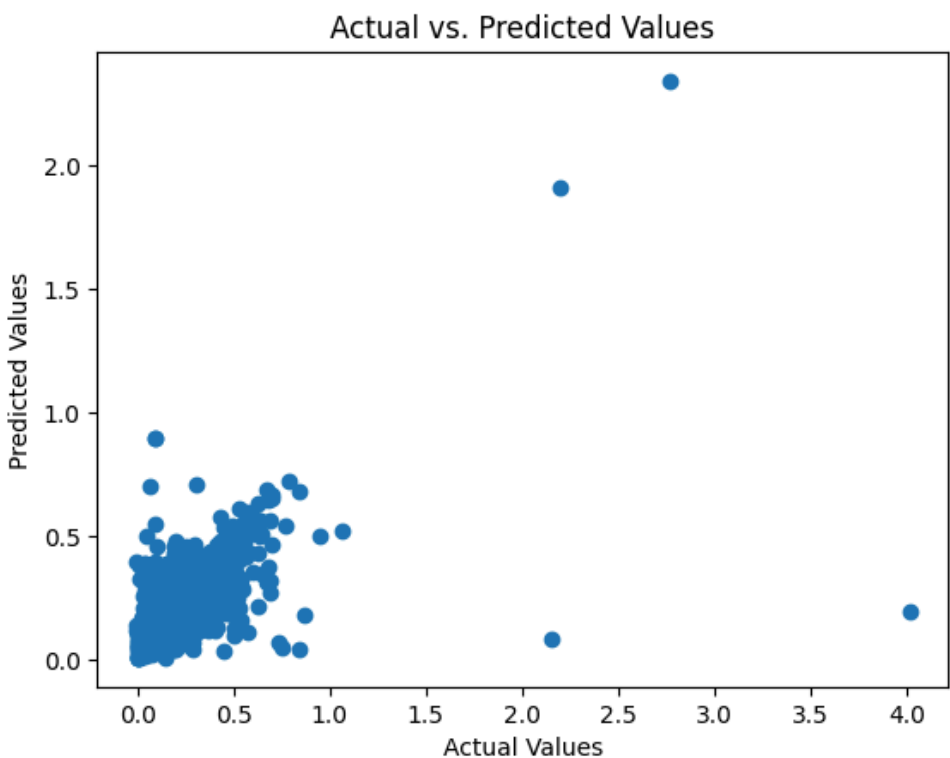
Confussion matrix of MSFT train data

Projects

Photometric Redshift Estimation Using Regression

github.com/Aldiansyah-ar/PhotometryRedshiftRegression

Redshift estimation using spectroscopic methods is constrained by high time consumption and computational costs. To overcome these limitations, a regression approach was implemented utilizing Extra Trees Regressor, Random Forest, and CNN algorithms. To optimize model performance, hyperparameter tuning was conducted using a genetic algorithm and random search.



Actual and predicted of redshift values

Model	Row Number (Data)	R ²	MSE	MAE
Deep Convolution Neural Network	237.269	0.74	1.4 x 10 ⁻³	2 x 10 ⁻²
Random Forest	237.269	0.79	1.1 x 10 ⁻³	1.6 x 10 ⁻²
Extra Tree (Random Search)	237.269	0.8	1.1 x 10 ⁻³	1.4 x 10 ⁻²
Extra Tree (GA)	237.269	0.79	1.1 x 10 ⁻³	1.62 x 10 ⁻²
Ensemble	237.269	0.81	1.0 x 10 ⁻³	1.5 x 10 ⁻²

Model comparison